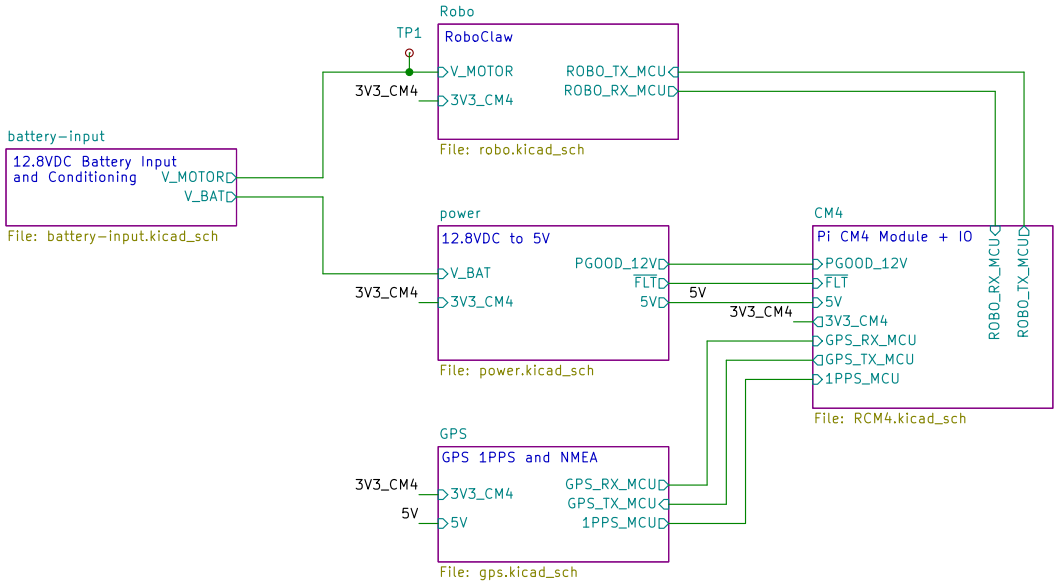
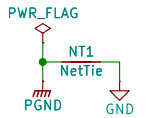
Raspberry Pi CM4 + RoboClaw Carrier Board (ROBOCLAW-2X15A-OEM)

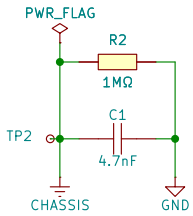


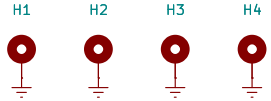
Motor Notes

File: motor-notes.kicad_sch

Grounding







NT1: single-point PGND-GND tie. Place at J1-2 (battery -). Motor current (RoboClaw BAT-) stays on PGND.

R2/C1: CHASSIS-GND bleed. Place near USB-C / USB-A / RJ45. Shields and RJ45 termination (P10) go to CHASSIS.

H1/H2/H3/H4: M3 plated mounting holes on CHASSIS. Bond to the metal enclosure or dish frame.

Sheet: /
File: cm4-robo-controller.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/12

CM4

Module1A
ComputeModule4-CM4
Hirose
2x DF40C-100DS-0.4V

Suggest a CM4004008.
4GB RAM, 8GB eMMC program storage.

Module1B
ComputeModule4-CM4
Hirose
2x DF40C-100DS-0.4V

GPIO

NB SD signals are only available on modules without eMMC

High Speed Serial

Module1 pins 89 (WiFi_nDisable) and 91 (BT_nDisable) are left unconnected on purpose. This board needs a CM4 WITHOUT wireless, where both pins are reserved (CM4 datasheet 2.1.1 / 2.1.2). If a wireless CM4 is ever fitted, disable the radios in config.txt: dtoverlay=disable-wifi and dtoverlay=disable-bt (the latter also frees UART0).

1PPS from GPS

File: usb.kicad_sch
ethernet

File: ethernet.kicad_sch
CM4_Connectors

CM4 power-health LED (PI_LED_nPWR, buffered per CM4 datasheet)

From the CM4 Manual: Active-low output to drive Power On LED. This signal needs to be buffered.

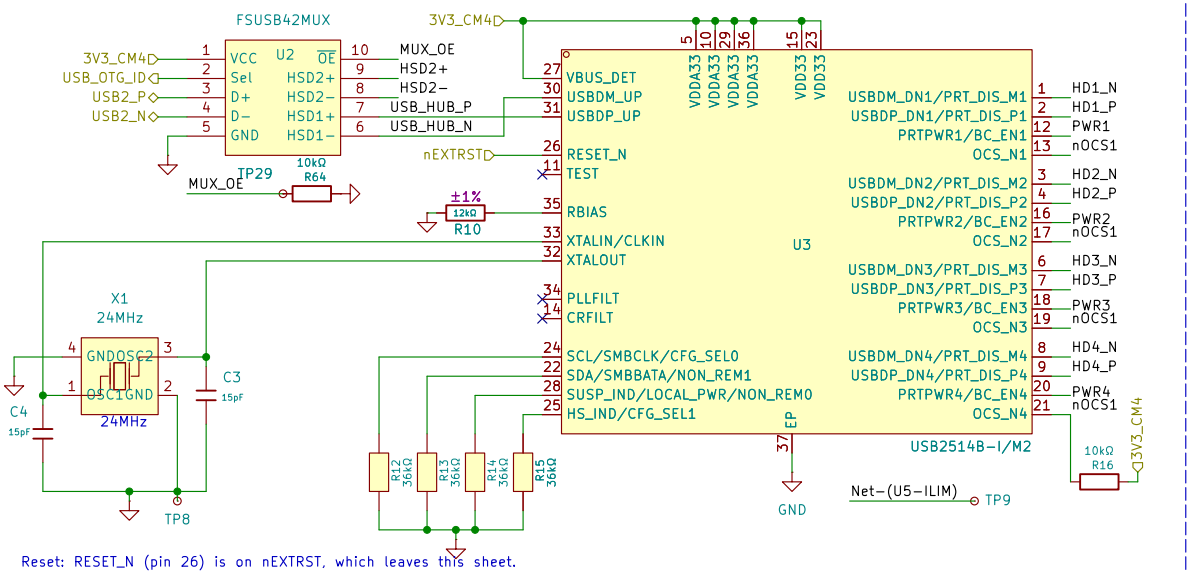
74LVC1G07SE-7

E6C0805UR

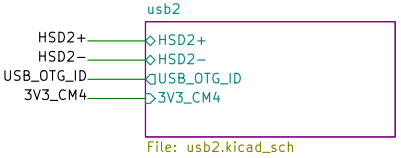
Sheet: /CM4/	
File: RCM4.kicad_sch	
Title:	
Size: A4	Date:
KiCad E.D.A. 10.0.1	Rev: Id: 2/12

USB – 2 Ports + 2 Port Expansion Header

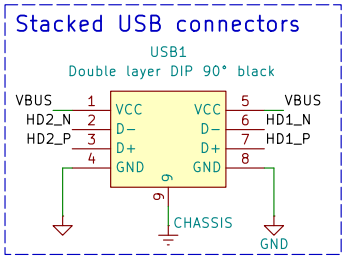
4-Port USB Hub (all 4 ports used)

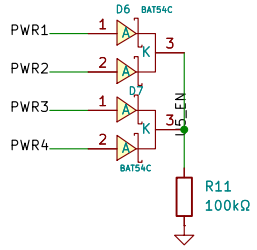


Reset: RESET_N (pin 26) is on nEXTRST, which leaves this sheet.



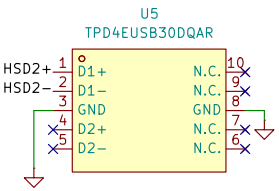
Stacked USB connectors



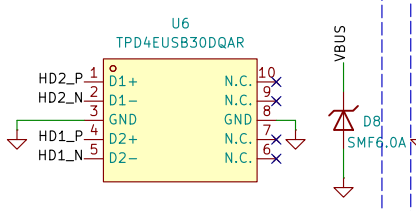


U4 port power, diode-OR (D6/D7, R11). The hub straps CFG_SEL=00: individual port power switching and OC sensing (USB2514B Table 5-1). U4 feeds all four ports, so its EN is the OR of PRTPWR1-4: VBUS stays on while any port is enabled and drops only when all four are off. OCS_N1-4 share nOCS1: a U4 fault is reported on every port (correct for one shared switch). BC_EN straps read 0 via the internal 10k IPD.

USB-C ESD

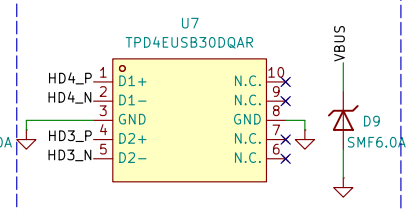


USB-A ESD



Protects both USB-A ports (4 lines).
Diode clamps VBUS. Place both at USB1.

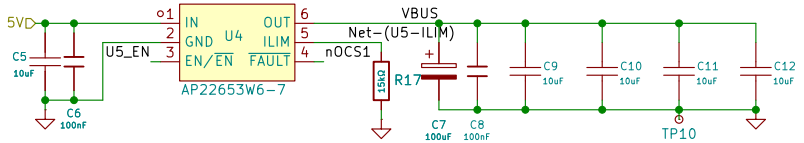
Expansion USB ESD



Protects both expansion ports (4 lines).
Diode clamps VBUS. Place both at J2.

Place ESD protection next to respective USB ports

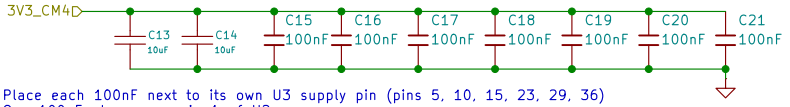
Current-limited high-side switch



Supplies and protects the 5V going out to all four USB ports
ILIM with 15K resistor limits to ~1.75A shared across USB1 and J2

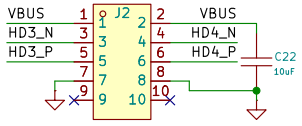
Expansion Port Note:
Pinout matches the standard PC front-panel USB 2.0 header.
Pin 9 is the key position – clip the post after reflow so
a keyed cable seats. Pin 10 NC.

3.3V Caps for U2 and U3 VDD/VCC.



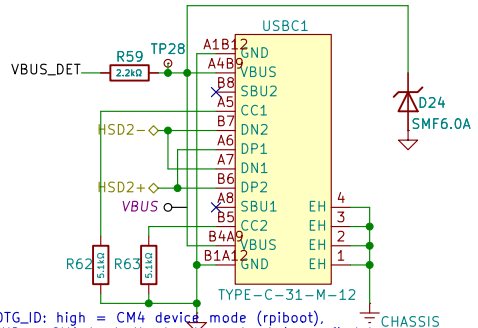
Place each 100nF next to its own U3 supply pin (pins 5, 10, 15, 23, 29, 36)
One 100nF place near pin 1 of U2

2 Port Expansion USB connector



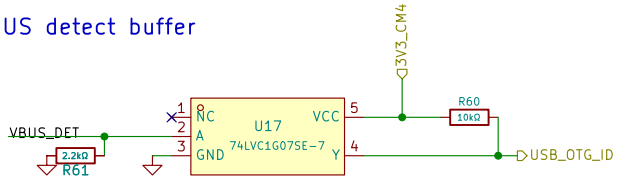
Sheet: /CM4/usb/		
File: usb.kicad_sch		
Title: USB		
Size: A4	Date: 2026-09-20	**Rev: 1.1**
KiCad E.D.A. 10.0.1		Id: 3/12

USB-C Port for CM4 Flashing (rpiboot)



USB_OTG_ID: high = CM4 device mode (rpiboot), low/GND = CM4 host. Needs otg_mode=1 in config.txt.
USBC1 VBUS: 5.0 V nom (4.75 V min / 5.5 V max).
No USB-PD controller; CC has 5.1 kΩ Rd only.
VBUS detect: R59/R61 (2.75 V max) -> U17 open-drain -> USB_OTG_ID.
VBUS current: 1.25 mA continuous / 1.25 mA peak.
VBUS survives 6.0 V DC (D24 SMF6.0A stand-off), surges clamped to 10.3 V; −0.7 V (D24 forward). U17 A sees half via R59/R61.

VBUS detect buffer



U17 is open-drain, powered from 3V3_CM4. With the CM4 off, Y is high-Z and R60 is unpowered, so U2 Sel and CM4 USB_OTG_ID are never driven (CM4 datasheet 5.1). 74LVC Ioff: A tolerates VBUS_DET at VCC = 0.

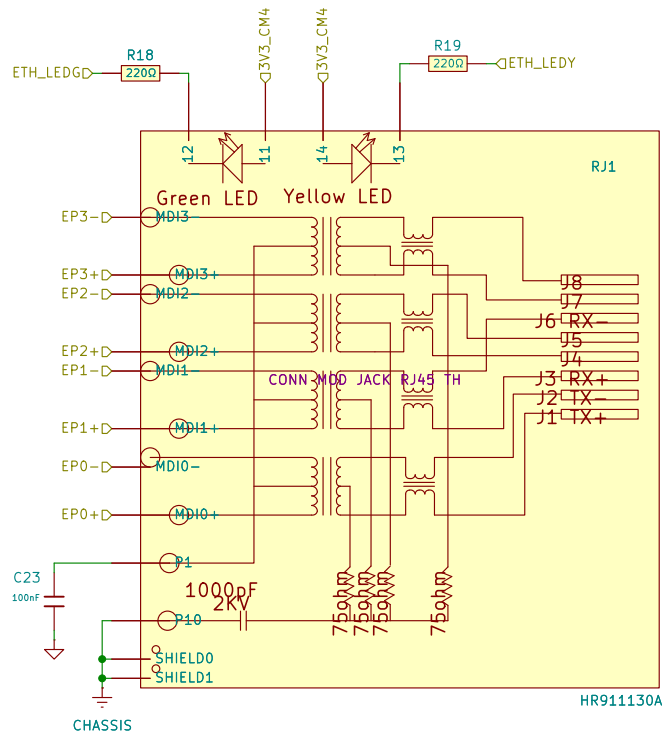
Sheet: /CM4/usb/usb2/
File: usb2.kicad_sch

Title: USB-C rpiboot port

Size: A4 | Date: 2026-10-13 | **Rev: 1.1**
KiCad E.D.A. 10.0.1 | Id: 10/12

Gigabit Ethernet — CM4 MDI to RJ45

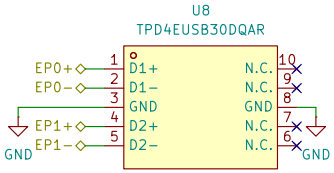
RJ45 with integrated magnetics

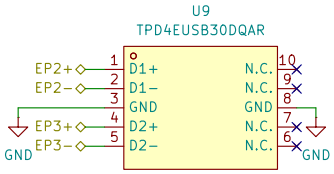


HR911130A, shielded, integrated magnetics and LEDs, 100/1000Base-T (LCSC C54408).
EP0–EP3 come straight from the CM4 — the PHY is on the module, no external PHY.
Chip-side center taps (P1) bypassed to GND by C23 100nF.
Bob Smith termination (4x75R + 1000pF/2kV) is built into the jack;
P10 is that node, tied to CHASSIS. Shield pins are also on CHASSIS
(bled to GND by R2/C1 on the top sheet).

CM4 Ethernet_nLED pins are active-low 3.3V sinks, IOL 8mA @ VOL 0.4V.
ETH_LEDG = activity (green, RJ1 pin 11 anode / 12 cathode).
ETH_LEDY = link speed (yellow, pin 14 anode / 13 cathode).
R18/R19 220R gives ~4.5mA through the LED. That is below the 8mA limit.
Route EP0–EP3 as 100R differential pairs, matched within 0.15mm.
Keep the RJ1 chassis/shield area clear of the signal ground pour.

Ethernet ESD





Clamps all four pairs on the CM4 side of the magnetics.
Place both at RJ1.

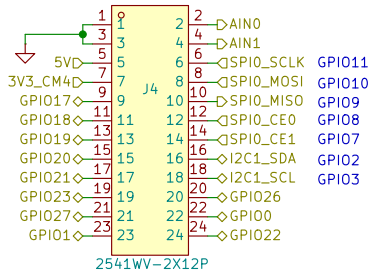
Matches CM4 datasheet Fig. 2 (TPD4EUSB30 on all pairs).
TPD4EUSB30 IO abs max is 0 to 6 V (TI p.4), so EP lines must stay
>= 0 V. MDI DC level is set inside the BCM54210PE
(not published; CT is AC-grounded only).
First boards: scope each EP line to GND at the clamps
at 1000/100/10 (ethtool -s eth0 speed N duplex full
autoneg off). Minimum must be >= 0 V (no undershoot below GND).

Use JLC's impedance calculator for the stackup to get
width and gap for 100 Ω (Ethernet) and 90 Ω (USB) differential
on the outer layers. Put those values into ETH_100_OHM and USB_90_OHM.
Order with impedance control enabled, so JLC adjusts to the target.

Sheet: /CM4/ethernet/ File: ethernet.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 4/12

CM4 External Header Pins

GPIO Headers



GPIO0 / GPIO1 (J4-22 / J4-23)
CM4 pins 36 (ID_SD = GPIO0) and 35 (ID_SC = GPIO1). This is the I2C0 bus the firmware normally uses to look for a HAT EEPROM and to control CSI0/DSI0 cameras and displays. This board has no HAT, camera or display, so the pins are free for general use.

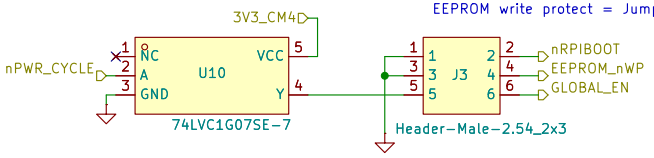
To use them as GPIO, add to config.txt (CM4 datasheet section 2.10):
force_eeprom_read=0
disable_poe_fan=1
Without these, the firmware drives the bus at boot to look for an EEPROM, and anything wired to these pins sees I2C activity.

- Both pins are pulled up (high) at reset.
- 3.3 V logic. Not 5 V tolerant.
- They can also be used as an I2C bus (I2C0: GPIO0 = SDA, GPIO1 = SCL). Add external pull-ups if you do.

AIN0 / AIN1 (J4-2 / J4-4), field wired, to CM4 AnalogIP0/1 (MxL7704 ADC). Measures 0 to 2.55 V (8-bit). Survives +/-24 V either polarity: R7/R6 10k series + D4/D5 BAT54S clamps to GND and 3V3_CM4 (RCM4 sheet) hold the pins within -0.3 V / +3.6 V (abs max -0.3/+6.0). Error: ADC input < 1 uA x 10k = 10 mV, plus clamp leakage (up to ~1-2 % hot). With the CM4 off, a driven input feeds < 0.3 mA (at 6 V) into 3V3_CM4 through the upper clamp.

Boot/EEPROM_EN/GLOBAL_EN

Force RPIBOOT = Jumper Pins 1-2
EEPROM write protect = Jumper Pins 3-4



POWER-CYCLE CONTROL (nPWR_CYCLE)
Lets the CM4 cut its own power after a clean shutdown and restart by itself. Use it when a normal reboot is not enough: it fully resets the PMIC and all on-module power rails.

How it works:
- GPIO6 drives U10 (74LVC1G07, open-drain). J3 pins 5-6 connect U10's output to GLOBAL_EN (CM4 pin 99, 100k internal pull-up to 5V).
- GPIO6 high (default pull-up at reset): U10 output is off and the CM4 runs.
- GPIO6 low: GLOBAL_EN is pulled low and the CM4 powers off.
- U10 is powered from 3V3_CM4, which switches off with the CM4, so U10 releases GLOBAL_EN and the CM4 powers back on.

Use:
- Fit the J3 5-6 jumper to enable. Fit J3 5-6 only after the board and GPIO6 behaviour have been verified.
- config.txt: dtoverlay=gpio-poweroff,gpiopin=6,active_low=1
"sudo poweroff" then shuts down cleanly and power-cycles the CM4.
- Do not drive GPIO6 low while the OS is running. CM4 datasheet section 2.13: pull GLOBAL_EN low only after the OS has shut down.
- The datasheet needs GLOBAL_EN low for >1 ms. The low time here is set by how fast 3V3_CM4 decays and is not guaranteed: check it with a scope on first boards.

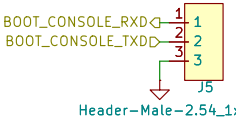
EEPROM WRITE PROTECT (EEPROM_nWP, J3 3-4)
The CM4 pulls EEPROM_nWP up internally (100k), so the bootloader EEPROM is writable unless J3 3-4 is fitted. The jumper only takes effect once protection is enabled in the bootloader config (CM4 datasheet 2.18).

Production:
- Set eeprom_write_protect=1 in the bootloader config (via usbboot or rpi-eeprom-config) before shipping.
- Fit the J3 3-4 jumper. The EEPROM is then protected from power-up (CM4 datasheet 5.1).

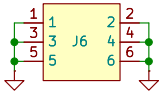
Bootloader update:
- Remove J3 3-4, update via usbboot or self-update, then refit the jumper.

Why not a GPIO with a pull-down: CM4 EEPROM updates run only via usbboot or bootloader self-update (datasheet 6.3), before Linux can drive a GPIO, so the pin could never be released for an update.

Boot Console UART



Gnd Pins



I2C PULL-UPS
I2C1: GPIO2 = I2C1_SDA (J4-16), GPIO3 = I2C1_SCL (J4-18) Pulled up to 3.3 V (GPIO_VREF) by 1.8 kΩ inside the CM4 (CM4 datasheet section 2.5, pins 56/58). No board pull-ups are needed. R20/R21 (2.2k, fitted via H5/H6) are optional.

I2C0: GPIO0 = ID_SD (J4-22), GPIO1 = ID_SC (J4-23)
No I2C pull-ups on the CM4, only the BCM2711's weak reset pull-ups (33-73 kΩ). If J4-22/23 are used as an I2C bus, the device on J4 must supply the pull-ups (2.2k-4.7k to 3.3 V).

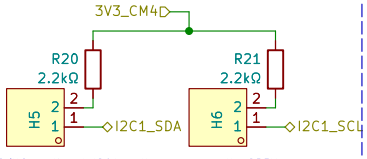
BOOT CONSOLE UART (J5): CM4 UART0, 3.3 V logic, 115200 baud, 8N1, no flow control
J5-1 BOOT_CONSOLE_RXD (GPIO15, CM4 receives) <- adapter TXD
J5-2 BOOT_CONSOLE_TXD (GPIO14, CM4 transmits) -> adapter RXD
J5-3 GND -- adapter GND
TX and RX cross over: the adapter TXD goes to the CM4 RXD.

Use an inexpensive USB-to-TTL serial cable (PL2303 / CP2102 / FTDI type) with female jumper leads. Typical colours: black GND, white RXD, green TXD, red 5V. Check the labels on your cable.

DO NOT connect the adapter 5V lead. The board powers itself and J5 has no 5V pin. A loose 5V lead can short to the header or the board, so it is best to cut it off (or insulate it).
Use an adapter with 3.3 V signals only. CM4 GPIOs are not 5 V tolerant: check that the adapter TXD idles at about 3.3 V with a meter before connecting.

Linux console: enable_uart=1 in config.txt (Pi OS cmdline.txt already has console=serial0,115200).
Bootloader messages: set BOOT_UART=1 in the EEPROM config (rpi-eeprom-config).

I2C Pull-Ups



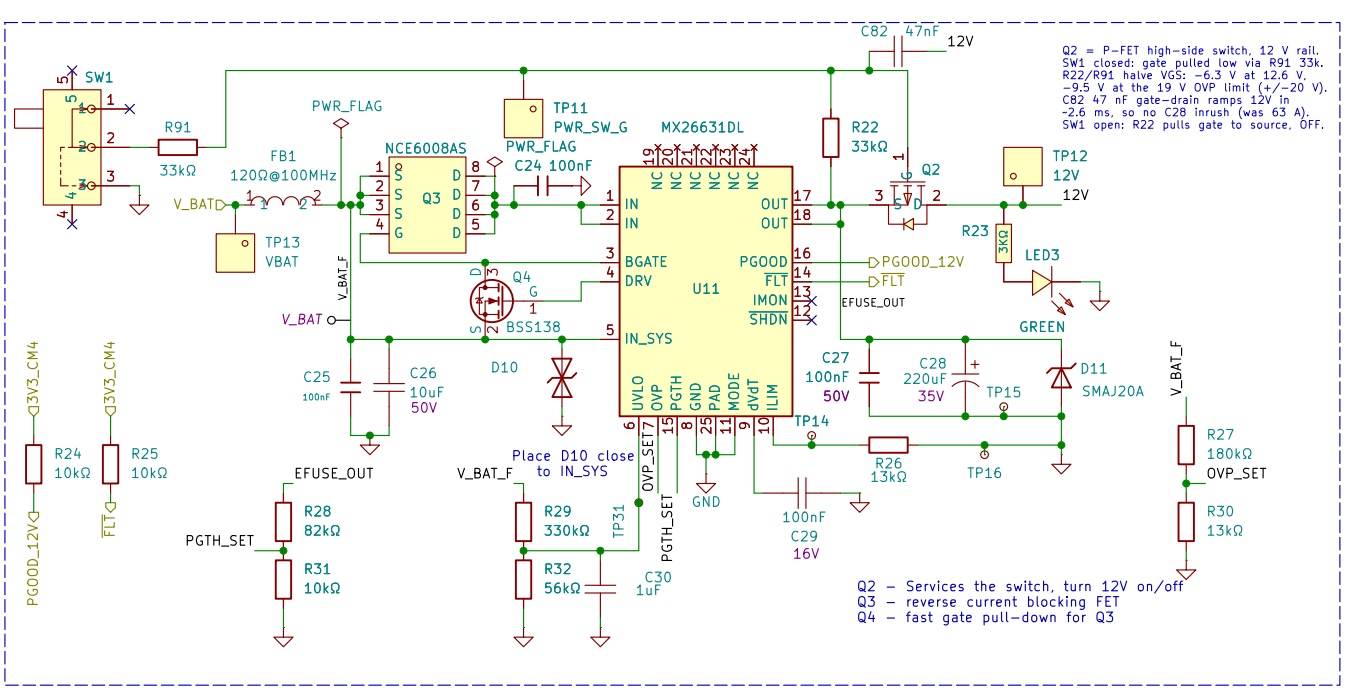
H5/H6: optional I2C1 pull-ups, normally OPEN. CM4 already pulls GPIO2/3 up with 1.8k (DS p.7). Fit both only for a long, high-capacitance bus at 400 kHz: ~1k total, ~2.9 mA sink at 0.4 V (I2C spec min 3 mA, so compliant devices are OK).

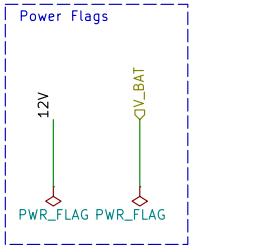
Sheet: /CM4/CM4_Connectors/
File: cm4_connectors.kicad_sch

Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 5/12

BATT 12V In to 12V
Current limiting, On/OFF switch, Reverse polarity protection





This circuit takes 12V from the battery, VBAT.
Reverse battery is blocked by Q1 at J1 (top sheet).
The eFuse limits to 3.08A typ (2.7-3.4A, see below).

EFUSE CURRENT LIMIT: R26 13k -> IOL = 40/13k = 3.08 A typ. Est. 2.7-3.4 A, from the datasheet IOL limits at 20k (-12/+8%) and 7.5k (-10/+11%) plus R26 1%. Branch demand ~1.9 A at 10 V: >= 40% margin at the low corner.

REVERSE BATTERY: blocked by Q1 (SP40P04NK P-FET) at J1 on the top sheet, ahead of the F2/F1 split. Reversed: VGS = 0 and the body diode is reverse-biased, so no current flows, no fuse blows, and V_BAT/V_BAT_F never go negative. U11 IN_SYS/UVLO/OVP/BGATE (abs min -0.3 V) stay within rating. D2 (15 V) clamps Q1 VGS during TVS clamping (~29 V). Q1 conducts both ways when on, so RoboClaw regen still reaches the battery. Q3 keeps its reverse-current-blocking role on this branch.
F2/F1 are Keystone 3557-2 holders (ATO fuses fitted at final assembly).

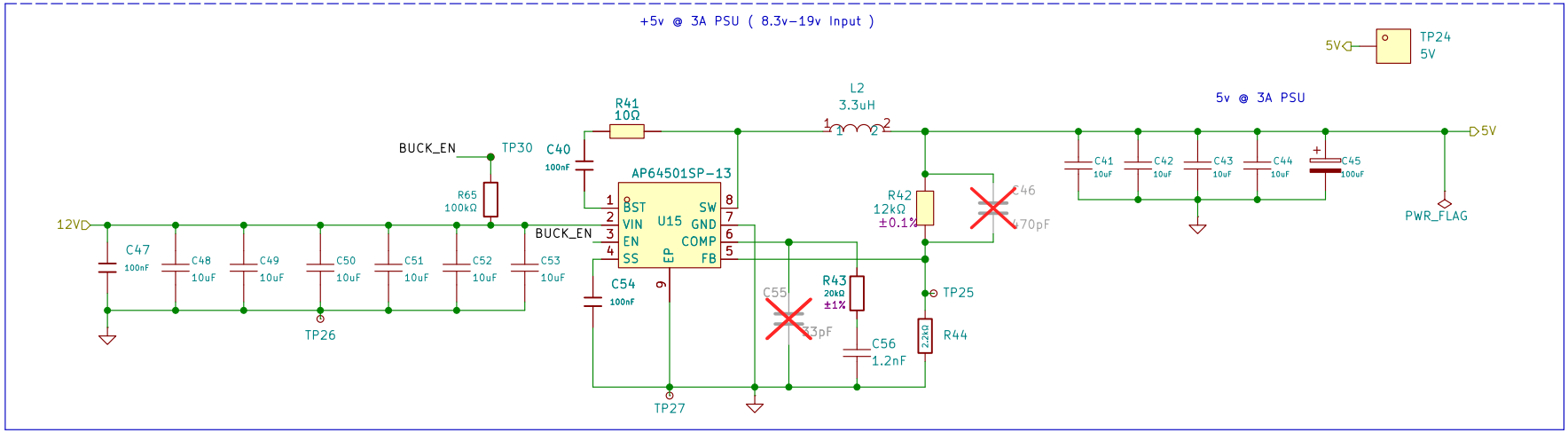
SW1 SWITCHES THE CM4 / LOGIC SUPPLY ONLY.
With SW1 off the board still draws about 31 mA from the battery: RoboClaw logic ~30 mA (fed from V_MOTOR), U11 + UVLO/OVP/PGOOD dividers ~1.3 mA. A full 12 Ah battery runs flat in about 16 days. For storage or transport, unplug J1 (or pull F1 and F2), and charge the battery first.

JRSeti
Sheet: /power/
File: power.kicad_sch

Title: Power

Size: A4	Date: 2026-08-18	**Rev: 2.0**
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12V to 5V



12V to 5V Buck – AP64501 (5A, 570kHz, 0.8V reference)
Vout = 5.16V, designed for 3A. Input 8.3V–19V (eFuse UVLO cuts in at 8.3V, OVP cuts out at 17.5–19V).
EN pulled up to VIN by R65 (100k): starts automatically, the eFuse upstream provides
the UVLO. TP30 on EN lets a test fixture hold the buck off.

Output voltage set by the FB divider

Vout = 0.8V x (1 + R42/R44) = 0.8 x (1 + 12k/2.2k) = 5.16V
Corners 5.10–5.22V (Vref 792–808mV, R42/R44 0.1%). Worst-case USB VBUS
= 5.10V – 40mV load reg – 1.908A x 135mΩ (U4) = 4.81V >= 4.75V;
max 5.22V stays within the CM4 5.25V limit. 1% resistors cannot meet both.
R42 12k 0.1% thin film top leg, 5V to FB
R44 2.2k 0.1% thin film bottom leg, FB to GND
C46 DNP 470pF optional feedforward across R42 for transient tuning
(datasheet example 47pF across 115k, scaled for R42 = 12k)

Loop compensation

R43 20k + C56 1.2nF series RC on COMP sets the loop response
C55 DNP 33pF optional HF pole (datasheet Table 1, 5V), fit only if needed

Gate drive and soft-start

C40 100nF bootstrap cap, BST to SW – powers the high-side FET driver
R41 10R boot resistor, slows the SW rising edge to cut EMI
C54 100nF soft-start; the internal 4uA source gives a ~20ms ramp

Input bulk and bypass

C48–C53 6 x 10uF 35V bulk; carries ~1.5A RMS ripple at 3A out
C47 0.1uF HF bypass, place closest to VIN/GND

Output filter

L2 3.3uH SRP7028A-3R3M Bourns molded, 6A Irms, 12A Isat, 30mΩ DCR
with 570kHz gives 1.56A p-p ripple at 12V in
(2.0A p-p at 19V in; peak coil current 4.0A at 3A load)
Isat 12A > U15 peak current limit (9.2A max), so L2
does not saturate during overload before hiccup
C41–C44 4 x 10uF 10V ceramic, low ESR
C45 100uF 6.3V polymer bulk for load steps; 5.16V = 82% of rating

Sheet: /power/Power_5V/
File: power_5v.kicad_sch

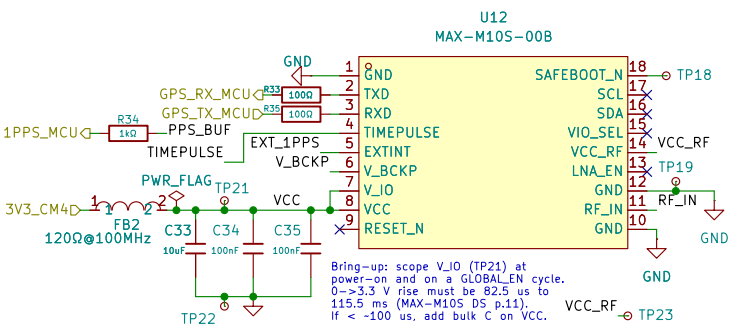
Title:

Size: A4 | Date: | **Rev:**
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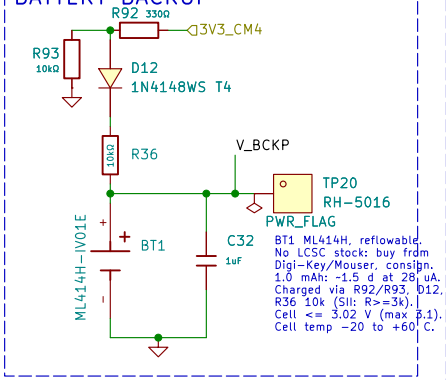
GPS

GPS is input using a MAX-M10S-00B.
NMEA sentences are used to get the UTC datetime and
the 1PPS is used to correct the MCU clock every second.
See https://www.mouser.com/pdfDocs/MAX-M10S_IntegrationManual_UBX-20053088.pdf

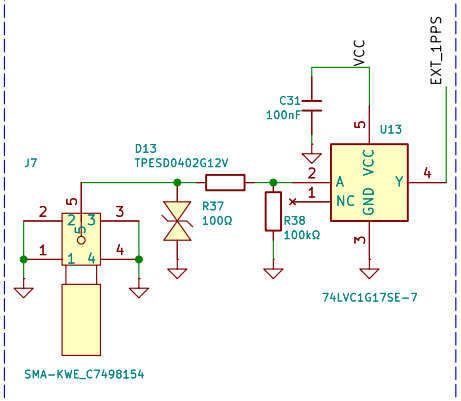
Net-(LED2-A) TP17



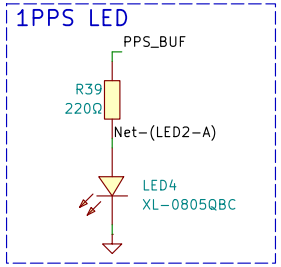
BATTERY BACKUP



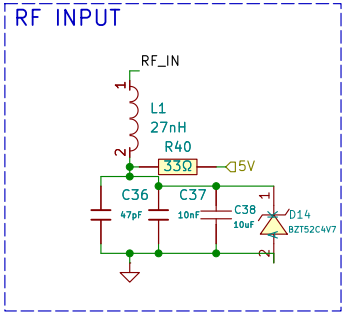
1PPS EXT INPUT (FE-5680A)



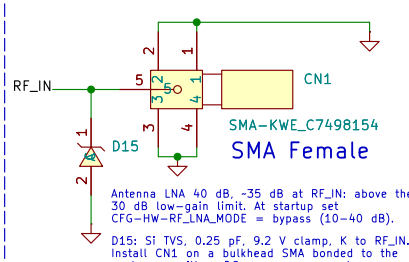
1PPS LED



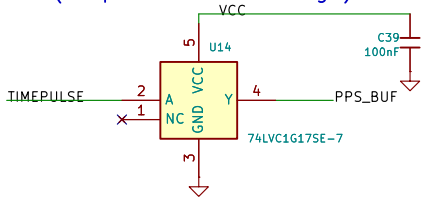
RF INPUT



RF SMA Connector



1PPS buffer (keeps SAFEBOOT_N high)



L2 supply-bead DCR must be < 0.2 Ω (u-blox MAX-M10S IM §3.1.1: series resistance on VCC causes ripple under dynamic current). BLM18KG121TN1D: 0.030 Ω max (0.040 Ω after test). Do not substitute a higher-DCR bead.

Antenna bias from 5V, not VCC_RF: the Waveshare L1/L2/L5 antenna needs 3.0–18 V at <=45 mA over 5 m RG174; VCC_RF (VCC-0.1 V, 50 mA max) cannot hold 3.0 V through any short-circuit resistor. R40 33R 1W limits a short to ~155 mA (L1 350 mA). C38 filters buck ripple. D14 4.7 V keeps RF_IN < 5.5 V DC abs max (antenna unplugged). Worst-case antenna voltage ~3.3 V at 45 mA.

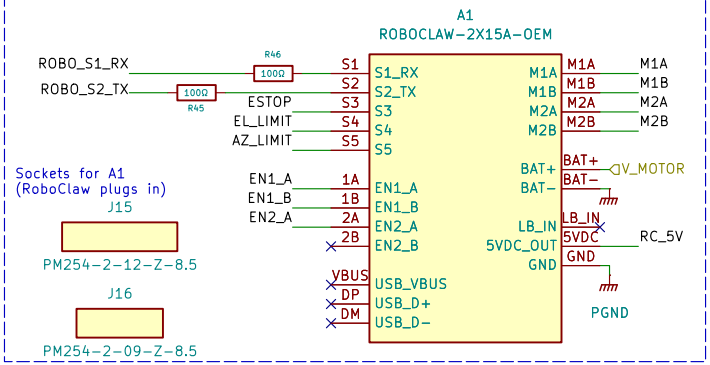
Sheet: /GPS/
File: gps.kicad_sch

Title: GPS 1PPS and RS232

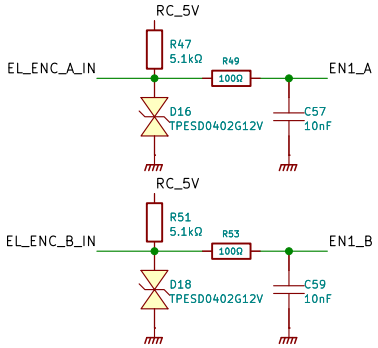
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ROBOCLAW-2X15A-OEM

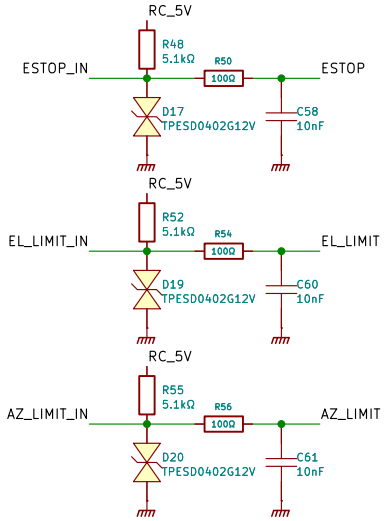
RoboClaw



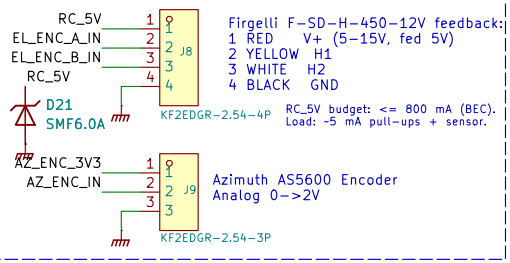
EN1: elevation encoder (Firgelli Hall, quadrature)



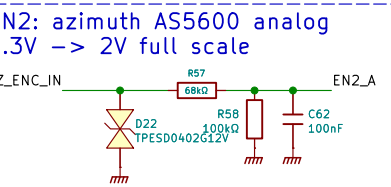
E-stop and limit inputs: NC switches, set Invert (active-high)



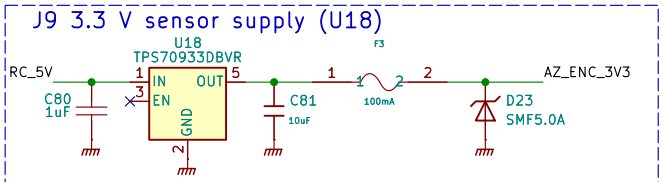
Encoder Inputs



EN2: azimuth AS5600 analog 3.3V -> 2V full scale

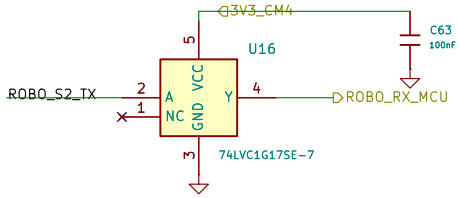


J9 3.3 V sensor supply (U18)



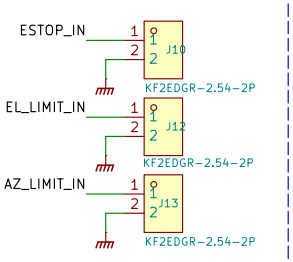
J9 (azimuth encoder, field-wired) has its own 3.3 V LDO, U18 (TPS70933, 30 V max, from RC_5V), not 3V3_CM4, so a miswired supply on J9 cannot back-feed the CM4 or the GPS. U18 blocks reverse current, so it cannot reach RC_5V either. EN is left open (internal pull-up; never tie EN to IN). F3 (100 mA PTC) limits a shorted cable, D23 (SMF5.0A) clamps J9. C81 >= 1.5 uF effective.

A1 TX buffer (no CM4 back-power)

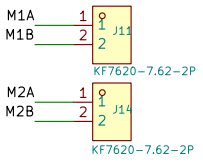


U16 isolates A1 S2 TX from CM4 GPIO13 (ROBO_RX_MCU); U19 (below the J9 box) isolates CM4 GPIO12 (ROBO_TX_MCU) from A1 S1 RX and its pull-up. A1 runs from the battery while the CM4 can be off (SW1 open, GLOBAL_EN power cycle). Wired direct, A1 would back-power the CM4 through GPIO13 or GPIO12 (abs max VREF+0.5 V). Both 74LVC1G17 run on 3V3_CM4; with it off, Ioff makes Y high-Z and A tolerates 3.3 V.

E-stop / limit switches



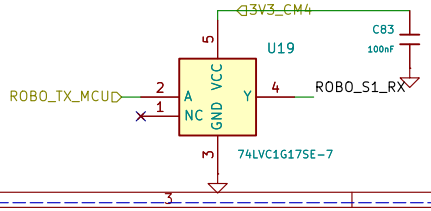
Motors Output
M1 = elevation
M2 = azimuth



J11 = Firgelli motor (elevation):
1 BLUE (+)
2 BROWN (-)
Swap to reverse, or
invert M1 in Motion Studio.

EMC: run motor leads in shielded cable, shield clamped 360 deg to the enclosure at the gland (not to J11/J14). Clip-on CM ferrite per pair at entry; 100 nF brush caps at the motors. Keep motor leads away from GNSS/encoder wiring.

A1 RX buffer (no CM4 back-power)



Sheet: /Robo/
File: robo.kicad_sch

Title: RoboClaw 2x15A Motor Controller

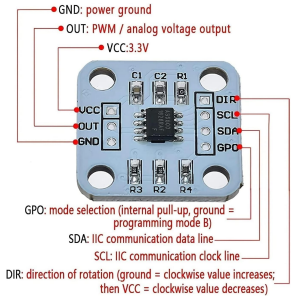
Size: A4	Date:	Rev:
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Notes About Motors + Axis Position Sensors

Azimuth Axis



Bringsmart JGY-370 Worm Gear Electric Motor 12V DC. Max 1.8A.



Typical azimuth encoder, AS5600

AZIMUTH SENSING: AS5600 analog absolute angle into RoboClaw EN2 (motor M2)

Wiring, J9 to A1 (RoboClaw):
J9-1 AZ_ENC_3V3 AS5600 supply: U18 3.3 V LDO (from RC_5V) via F3 (100 mA PTC), D23 clamp.
Run the AS5600 at 3.3 V (VDD5V tied to VDD3V3 on the sensor board).
J9-2 AZ_ENC_IN AS5600 OUT. D22 ESD clamp. R57 68k / R58 100k divider (x0.595) -> EN2_A (A1 2A).
C62 100nF filters it (40.5k x 100nF, tau = 4 ms, about 39 Hz).
J9-3 PGND Sensor return.
EN2_B is unused; the analog signal needs one channel only.
Scaling: AS5600 0-3.3 V -> 0-1.96 V at EN2_A, inside the RoboClaw 0-2 V analog input range.

How the RoboClaw uses it:
- Set EN2 to Absolute encoder mode in Motion Studio. The voltage is an absolute angle, so no homing is needed at power-up.
- M2 runs position PID on this reading. Speed is derived from the rate of change of position; there is no separate speed sensor.
- Read the RoboClaw position value at each end of travel and set those as the M2 min/max position limits.
- The AS5600 is single-turn: its output jumps at the 0/360 deg wrap. Fit the magnet so the azimuth travel never crosses the wrap, or set ZPOS/MPOS (I2C or OTP) to place the range. A crossing makes the position loop run away.
- The magnet must be on a shaft that turns at most one revolution over the full azimuth travel.

AS5600 output loading (datasheet): full-range analog (the default) needs >= 100k. R57 + R58 = 168k, OK.
- OUT capacitive load 1 nF max: R57 isolates C62; keep the J9 cable short (about 100 pF/m).
- Divider source impedance is about 40k. If the RoboClaw EN2 pull-up stays on in Absolute mode it lifts the reading: bench-check EN2_A with and without the RoboClaw; calibrate travel limits on hardware.
- J9 is powered from 3V3_CM4 (logic GND) but returns on PGND, so the sensor current crosses NT1.

Elevation Axis



Firgelli F-SD-H-450-12V actuator. Max 5.5A. 450 lbs force.
https://www.firgelliauto.com/products/super-duty-actuators
There are no limit switches.
Feedback from Hall sensors.

Future addition? Firgelli makes a version with adjustable limit switches:
https://www.firgelliauto.com/products/adjustable-stroke-linear-actuators?variant=39421954261063
But only up to 220 lbs force.

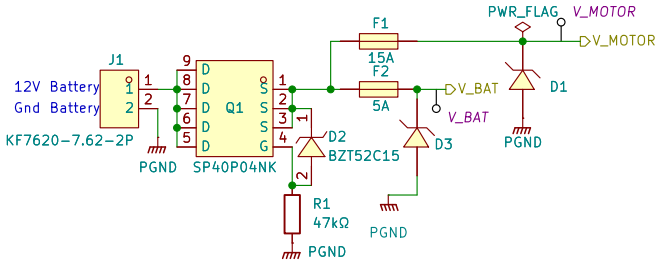
ELEVATION SENSING: Firgelli Hall quadrature into RoboClaw EN1 (motor M1)

Wiring, J8 to A1 (RoboClaw):
J8-1 RC_5V Hall sensor supply from the RoboClaw 5VDC output (800 mA max, shared).
D21 SMF6.0A clamps it.
J8-2 EL_ENC_A_IN Hall A. R47 5.1k pull-up to RC_5V, D16 ESD clamp, R49 100R + C57 10nF
-> EN1_A (A1 1A).
J8-3 EL_ENC_B_IN Hall B. R51 5.1k pull-up to RC_5V, D18 ESD clamp, R53 100R + C59 10nF
-> EN1_B (A1 1B).
J8-4 PGND Sensor return, on the RoboClaw ground.
The pull-ups suit open-collector Hall outputs and are harmless with push-pull outputs.
The 100R/10nF filters (tau = 1 us, about 160 kHz) pass the Hall pulses and remove cable noise and ringing.
RoboClaw inputs tolerate 5 V, so no level shifting is needed.

How the RoboClaw uses it:
- The two Hall sensors are 90 degrees apart (quadrature A/B). The RoboClaw counts the pulses, and the A/B phase gives the direction. If the count falls while the actuator extends, swap A/B in Motion Studio; no rewiring is needed.
- Position = encoder count. Speed = count rate. Set QPPS (counts/s at full speed) in Motion Studio, then auto-tune the M1 velocity and position PID.
- The encoder is incremental, so the count is lost at power-off and the axis must be homed at start-up. The actuator has no limit switches. Home on the elevation limit switch (J12 -> S4, set as home/limit), or drive slowly to full retraction with a low current limit, then zero the counter.
- Stroke = counts / (Firgelli pulses per inch). Elevation angle follows from the mount geometry, which is non-linear, so convert between angle and counts on the CM4.

Sheet: /Motor Notes/		
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12.8VDC Input Conditioning



Powered by a 12.8V 12Ah LiFePO4 Lithium battery IP67 Waterproof

V_MOTOR (A1 BAT+, via F1 15 A): motors 7.5 A max (F-SD-H-450 actuator 5.5 A + worm motor 2 A).
F2 5 A feeds the CM4 board branch (V_BAT) only.
F2/F1: Keystone 3557-2 holders; fit Littelfuse ATO 0287005.PXCN (5 A) / 0287015.PXCN (15 A).

V_BAT (J1): 12.8 V nom, 10.0 V min, 14.6 V max (charging).
Survives 18 V max (TVS). Reverse battery: Q1 P-FET blocks it; survives −14.6 V, nothing conducts, no fuse blows.

Q1 – 40V P-Channel Mosfet
Give Q1's drain tab a solid copper pour; it dissipates about 0.55 W worst case.

F1/F2 are the backup for a reversed battery: they blow only if Q1 has failed short (about 270 A then flows through D1/D3 and the RoboClaw diodes). Check polarity and Q1 before replacing a blown fuse.

Standby drain: SW1 (power sheet) switches the CM4 / logic supply only. With SW1 off the battery still feeds the RoboClaw logic (~30 mA, via F1) and U11 + dividers (~1.3 mA): about 31 mA, so 12 Ah lasts ~16 days. For storage or transport unplug J1 (or remove F1/F2); charge first.

JRSeti		
Sheet: /battery-input/		
File: battery-input.kicad_sch		
Title: Battery Input		
Size: A4	Date: 2026-10-17	**Rev: 2.0**
KiCad E.D.A. 10.0.1		Id: 17/12